

Bringing Adventure Gaming to Life Using Real-Time Generative AI on Your PC

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ABSTRACT

Imagine a new kind of tabletop gaming experience, where a narrator is describing a complex fantasy world, and players gathered around the table can see the events of their world unfolding in real-time, on their PC devices. In this talk, we will show attendees how to execute multi-modal Generative AI (Gen AI) models, running on a PC, in real-time to create immersive scenery, followed by an interactive live demonstration. This talk walks through the optimization of Gen AI modalities, chaining audio transcription and diffusion models, together in real-time. We compress these models with the OpenVINO™ Toolkit and leverage the Intel®Core™ Ultra processor to split these workloads across CPU, integrated GPU, and Neural Processing Unit (NPU), getting the best performance for each model. We also cover exciting new developments with temporal-consistent and depth estimation approaches towards high-resolution, 3D, pop-up, scenery generation. This talk equips participants with hands-on tools to resolve challenges with real-time, high-quality gaming scenery generation on their PC for gaming.

KEYWORDS

Deep Learning, Artificial Intelligence, Generative AI, AI Compute, Multi-modal Generative AI, 3D Scene Generation, Gaming

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1 INTRODUCTION

Generative AI (Gen AI) has profound effects on transforming gaming experiences into immersive, reactive environments. Gamers are looking for cost-effective setups that handle powerful game mechanics locally, on their PCs, without network lag and cloud costs. We present our workflow with Gen AI models for real-time, on-device audio transcription and image generation, producing high-quality varied images that match the desired aesthetics and gameplay of different types of table-top games, such as the style of Dungeons and Dragons or The Oregon Trail. We discuss challenges of executing Gen AI on PCs and solutions, including software techniques and intelligent offloading across CPU, GPU, and Neural Processing Unit (NPU) engines. Prior work [Santiago III et al., 2023; Callison-Burch et al., 2022; Triyason, 2023] explored AI in gaming setups and dialog agents, while we focus on Gen AI's interaction with storytellers for adventure gaming. The popularity of models generating video scenery (Lumiere [Bar-Tal et al., 2024] and Sora [Brooks et al., 2024]) also merit an exploration of approaches for gaming. This talk builds on our work on Gen AI optimization [Kozlov et al., 2021; Savelyev et al., 2024; Talamanova et al., 2023] and its application to gaming environments [Wu and Lo, 2023]. Our code is available on GitHub here.

2 IMAGINATIVE SCENERY WITH GEN AI

Scenery must be created instantly as a narrator speaks, with instant audio transcription and image generation, occurring on-device without the need for an internet connection, ensuring data remains local. We created a workflow (see Figure 1) integrating multiple state-of-the-art Gen AI and AI models running in sequence, on-device: (1) Whisper/Distil-Whisper [Ma et al. 2023; Gandhi et al., 2023] for audio transcription, (2) Latent Consistency Models [Luo et al., 2023] for image generation, (3) Super Resolution to upscale the result, (4) Depth Anything [Yang et al., 2024], to add a 3D “pop-up” effect for temporal-consistency. On **user experience**: The more descriptive the prompt, the higher the output quality,

allowing the narrator to articulate their creative vision. A post-processing mechanism and not safe for work (NSFW) filter ensure appropriate prompt structuring, while an optional Large Language Model (LLM) component infuses a prompt with a theme. Players have discretion to use the system during their session via Start/Stop buttons – future work includes a wake word for discreet generation. All models must be used in compliance with their licenses, with avoidance of copyright content.

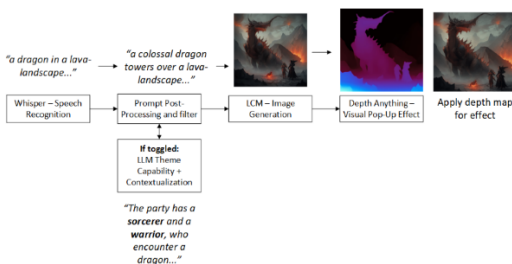


Figure 1: Workflow for scenery generation

Optimizations Gen AI models pose challenges due to their large runtime and model size. We apply optimizations to run these models on client machines, with compression strategies to reduce model size, footprint, and runtime. The size and footprint of Whisper [Ma et al. 2023], an automatic end-to-end speech recognition (ASR) model, are reduced by converting the models to lower precisions - from FP32 to FP16 and INT8 [Savelyev et al., 2024] – similar techniques are used for the LLM agent and Depth Anything models, in addition to model caching to significantly reduce compile time. Latent Consistency Models (LCM) are compressed to FP16 precision with OpenVINO™, halving the memory footprint and maintaining fast inference (see Figure 2 for details) [Talamanova et al., 2023]. Further optimization is achieved using OpenVINO™’s Neural Network Compression Framework (NNCF) for post-training quantization, involving adding quantization layers into the model graph and initializing these layers’ parameters using a subset of a training dataset. Executing these quantized operations in INT8 precision on GPU accelerates model inference vs FP32/FP16 variants. We additionally use HW accelerators to perform multiple execution with different engines to handle on-device processing demands: CPU for fast response, GPU for larger workloads, and the NPU for executing AI workloads with greater efficiency. By using the Intel®Core™ Ultra AI PC, we flexibly split components of our AI workflow across accelerators – e.g., splitting Whisper’s encoder and decoder across our NPU and GPU, or running variants of Whisper entirely on NPU. For compute-intensive workloads, we execute the LCM image-generation model directly on GPU/NPU. We will demonstrate the workflow-split by assigning components of the models to the right HW accelerators, with a live demo of the end-to-end storytelling experience; and conclude with our efforts on additional frame interpolation effects for immersive scenery, and the future potential of AI generation models for interactive gaming.



Figure 2: Reducing diffusion iterations and optimize LCM with NNCF and OpenVINO™ for inference speed.

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REFERENCES

- Jose Ma. Santiago III, Richard Lance Parayno, Jordan Aiko Deja, and Briane Paul V. Samson. 2023. Rolling the Dice: Imagining Generative AI as a Dungeons & Dragons Storytelling Companion. arXiv:2304.01860. Retrieved from <https://arxiv.org/abs/2304.01860>
- Chris Callison-Burch, Gaurav Singh Tomar, Lara Martin, Daphne Ippolito, Suma Bailis, and David Reitter. 2022. Dungeons and Dragons as a Dialog Challenge for Artificial Intelligence. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 9379–9393, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
- Tuul Triyason. 2023. Exploring the Potential of ChatGPT as a Dungeon Master in Dungeons & Dragons tabletop game. In *13th International Conference on Advances in Information Technology (IAIT 2023), December 06–09, 2023, Bangkok, Thailand*. ACM, New York, NY, USA, 6 Pages. <https://doi.org/10.1145/3628454.3628457>
- Omer Bar-Tal, Hila Chefer, Omer Tov, Charles Herrmann, Roni Paiss, Shiran Zada, Ariel Ephrat, Junhwa Hur, Guanghui Liu, Amit Raj, Yuanzhen Li, Michael Rubinstein, Tomer Michaeli, Oliver Wang, Deqing Sun, Tali Dekel, Inbar Mossieret. al. 2024. Lumiere: A Space-Time Diffusion Model for Video Generation. arXiv:2401.12945. Retrieved from <https://arxiv.org/abs/2401.12945>
- Tim Brooks, Bill Peebles, Connor Holmes, Will DePue, Yufei Guo, Li Jing, David Schnurr, Joe Taylor, Troy Luhman, Eric Luhman, Clarence Ng, Ricky Wang, Aditya Ramesh. 2024. Video generation models as world simulators. (Feb 2024). Retrieved Feb 21, 2024 from <https://openai.com/research/video-generation-models-as-world-simulators>
- Zhuo Wu and Raymond Lo. 2023. Connecting the Real-World and Metaverse via Real-Time Digitization with Intel Arc. In *SIGGRAPH Asia 2022 Courses (SA '22)*. Association for Computing Machinery, New York, NY, USA, Article 3, 1–14. <https://doi.org/10.1145/3550495.3558224>
- Simian Luo, Yiqin Tan, Longbo Huang, Jian Li, and Hang Zhao. 2023. Latent Consistency Models: Synthesizing High-Resolution Images with Few-Step Inference. arXiv:2310.04378. Retrieved from <https://arxiv.org/abs/2310.04378>
- Sanhit Gandhi, Patrick von Platen, and Alexander M. Rush. 2023. Distil-Whisper: Robust Knowledge Distillation via Large-Scale Pseudo Labelling. arXiv:2311.00430. Retrieved from <https://arxiv.org/abs/2311.00430>
- Alexander Kozlov, Ivan Lazarevich, Vasily Shamporov, Nikolay Lyalyushkin, and Yuri Gorbachev. (2021). Neural Network Compression Framework for Fast Model Inference. In: Arai, K. (eds) *Intelligent Computing. Lecture Notes in Networks and Systems*, vol 285. Springer, Cham. https://doi.org/10.1007/978-3-030-80129-8_17
- Nikita Savelyev, Alexander Kozlov, Ekaterina Aidova, and Maxim Proshin. "Optimizing Whisper and Distil-Whisper for Speech Recognition with OpenVINO™ and NNCF," Intel Corporation, 29 Jan. 2024. Available online: Intel Community Blog. [Accessed 19 February 2024].
- Liubov Talamanova, Ekaterina Aidova, and Alexander Kozlov. "Optimizing Latent Consistency Model for Image Generation with OpenVINO™ and NNCF," Intel Corporation, 23 Nov. 2023. Available online: Intel Community Blog. [Accessed 19 February 2024].
- Lihe Yang, Bingyi Kang, Zilong Huang, Xiaogang Xu, Jiashi Feng, and Hengshuang Zhao. 2024. Depth Anything: Unleashing the Power of Large-Scale Unlabeled Data. arXiv:2401.10891. Retrieved from <https://arxiv.org/abs/2401.10891>